

IMO 2026 — Problem 6

Solution by Kimi K3

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Source: `results/kimi-k3/problem-06/current.md` — autonomous run, no human intervention; solution as written by the model. Multi-round runs: earlier-round audit logs in `scratch/`.

Problem

Let $a_1, a_2, a_3, \dots$ be an infinite sequence of positive integers greater than 1. Suppose that for all positive integers n , the number a_{n+1} is the smallest positive integer greater than a_n such that $\gcd(a_{n+1}, a_i) > 1$ for every $i = 1, 2, \dots, n$. Prove that there exist positive integers T and L such that $a_{n+T} = a_n + L$ for every positive integer n . (Note that $\gcd(x, y)$ denotes the greatest common divisor of positive integers x and y .)

Solution document

Status

solved

Approaches tried

- **Minimal-hitting-set (transversal) framework** (previous session): $S_n = \{m : \gcd(m, a_i) > 1 \ \forall i \leq n\}$; $H_n =$ minimal transversals of $\{\text{supp}(a_1), \dots, \text{supp}(a_n)\}$; $a_{n+1} =$ successor of a_n in $S_n = \min_{h \in H_n}$ next multiple of h . Gap bound $a_{n+1} - a_n \leq \text{rad}(a_1)$; Lemmas A–D (min-value permanent exists; disjoint pairs die; pure-power terms; permanents = radicals of pure-power terms). Empirics: shrinkage of H_n always stops; stable H pairwise intersecting; $a_{n+T} = a_n + L$ verified for 100+ values of a_1 .
- **(THIS SESSION) Self-dual intersecting clutter — complete resolution.** The permanents (= minimal elements of S_∞) form a family $\mathcal{C}$ of finite prime-sets that is (i) an antichain, (ii) pairwise intersecting, (iii) self-dual: every minimal transversal of $\mathcal{C}$ is a member of $\mathcal{C}$. Then a purely combinatorial theorem (proved below) says: **any such family is finite**. Finiteness of permanents $S_\infty =$ multiples of finitely many permanents S_∞ periodic mod L $a_{n+T} = a_n + L$. This bypasses the old “shrinkage stops” crux entirely (Lemmas A–D, gap bounds, potentials are NOT needed for the final proof — only for intuition).

Current best

Complete proof assembled (see Full proof below). Structure: (1) the sequence is exactly the increasing enumeration of $S_\infty \cap [a_1, \infty)$; (2) if S_∞ has finitely many minimal elements, conclusion follows with $L = \text{lcm}$ of minimals, $T = \text{count per period}$; (3) the minimal elements (permanents) form an antichain of finite prime sets that is pairwise intersecting and self-dual; (4) **combinatorial core theorem**: any antichain of finite sets that is intersecting and self-dual is finite — proved by an infinite-descent/peeling process (transversal-minimal T inside a finite $I \cup Z_s$ would be a member properly contained in another member, contradicting the antichain property). Numerical checks of the framework and of permanents' intersecting+self-dual properties in `code/`.

Full proof

Notation. For an integer $m \geq 2$, $\text{supp}(m) = \text{set of prime divisors of } m$. For a finite set s of primes, $\pi(s) := \prod_{p \in s} p$ (a squarefree integer). A set s of primes is a *transversal* of a family $\mathcal{F}$ of sets if $s \cap F \neq \emptyset$ for every $F \in \mathcal{F}$.

Step 1: The sequence enumerates $S_\infty \cap [a_1, \infty)$

Let $S_\infty := \{m \geq 1 : \gcd(m, a_i) > 1 \text{ for all } i \geq 1\}$.

Claim. $(a_n)_{n \geq 1}$ is the increasing enumeration of $S_\infty \cap [a_1, \infty)$.

Proof. Every term lies in S_∞ : for $i < j$, the defining property of a_j (smallest integer $> a_{j-1}$ with $\gcd(a_j, a_{i'}) > 1$ for all $i' \leq j-1$) applied with $i' = i$ gives $\gcd(a_j, a_i) > 1$; also $\gcd(a_i, a_i) = a_i > 1$. So any two terms share a factor > 1 , i.e. every term is in S_∞ ; also $a_n \geq a_1$ as the sequence is strictly increasing.

Conversely let $x \in S_\infty$, $x \geq a_1$, and suppose x is not a term. Then $x > a_1$ (else $x = a_1$ is a term), and since the terms form an infinite strictly increasing sequence of integers, there is a unique n with $a_n < x < a_{n+1}$. By definition a_{n+1} is the smallest integer $> a_n$ in $S_n := \{m : \gcd(m, a_i) > 1 \forall i \leq n\}$. But $x \in S_\infty \subseteq S_n$ and $x > a_n$, so $a_{n+1} \leq x$ — contradiction. ■

Step 2: Reduction to finiteness of the minimal elements of S_∞

S_∞ is closed under taking multiples. Let $\mathcal{M}$ be the set of its minimal elements under divisibility. Every $m \in S_\infty$ is a multiple of some element of $\mathcal{M}$: among the divisors of m lying in S_∞ (a nonempty finite set) the smallest is a minimal element. Hence $S_\infty = \bigcup_{h \in \mathcal{M}} h\mathbb{Z}^+$.

Claim. If $\mathcal{M}$ is finite, the problem's conclusion holds.

Proof. Write $\mathcal{M} = \{h_1, \dots, h_k\}$ and $L := \text{lcm}(h_1, \dots, h_k) \geq 1$. Since each $h_i \mid L$, for every positive integer m : $m \in S_\infty \iff \text{some } h_i \mid m \iff \text{some } h_i \mid m + L \iff m + L \in S_\infty$. Let $T := \#(S_\infty \cap [a_1, a_1 + L))$; since $a_1 \in S_\infty$ (Step 1), $T \geq 1$. The map $x \mapsto x + L$ is an order-preserving bijection $S_\infty \cap [a_1, \infty) \rightarrow S_\infty \cap [a_1 + L, \infty)$; iterating, each block $[a_1 + jL, a_1 + (j+1)L)$ ($j \geq 0$) contains exactly T elements of S_∞ , namely $a_1 + jL, \dots, a_T + jL$. By Step 1 the terms in that block are exactly $a_{jT+1}, \dots, a_{jT+T}$, so $a_{jT+r} = a_r + jL$ for $1 \leq r \leq T$, $j \geq 0$. Writing $n = jT + r$: $a_{n+T} = a_{(j+1)T+r} = a_r + (j+1)L = a_n + L$ for every $n \geq 1$. ■

Step 3: The minimal elements form an intersecting, self-dual antichain of finite sets

Every $h \in \mathcal{M}$ is squarefree: if $p^2 \mid h$ for a prime p , then $h/p \in S_\infty$ (since $\text{supp}(h/p) = \text{supp}(h)$ still meets every $\text{supp}(a_i)$), contradicting minimality. So each $h \in \mathcal{M}$ has the form $h = \pi(t)$ for a finite set of primes $t = \text{supp}(h)$. For squarefree $h = \pi(t)$ we have $h \in S_\infty$ iff t is a transversal of $\{\text{supp}(a_i) : i \geq 1\}$, and h minimal under divisibility iff t is minimal under inclusion (a proper divisor of a squarefree h lying in S_∞ is exactly a proper sub-transversal). Hence

$$\mathcal{C} := \{\text{supp}(h) : h \in \mathcal{M}\}$$

is exactly the family of minimal (under inclusion) transversals of $\{\text{supp}(a_i) : i \geq 1\}$. It is an antichain of finite sets: if $t \subseteq t'$ are both minimal transversals, then $t = t'$.

Claim 3.1 (intersecting). Any two members of $\mathcal{C}$ intersect.

Proof. Let $t \in \mathcal{C}$. Every multiple of $\pi(t)$ lies in S_∞ (its support contains t , hence meets every $\text{supp}(a_i)$). Choose $k \geq 1$ with $\pi(t)^k \geq a_1$; by Step 1, $\pi(t)^k$ is a term a_j , and $\text{supp}(a_j) = t$. Every $t' \in \mathcal{C}$ is a transversal of $\{\text{supp}(a_i)\}$, so $t' \cap t = t' \cap \text{supp}(a_j) \neq \emptyset$. ■

Claim 3.2 (self-dual). Every minimal transversal of the family $\mathcal{C}$ belongs to $\mathcal{C}$.

Proof. Let T be a minimal (under inclusion) transversal of $\mathcal{C}$. For each i , $a_i \in S_\infty$ is a multiple of some $\pi(t)$ with $t \in \mathcal{C}$; then $t \subseteq \text{supp}(a_i)$, and T meeting t implies T meets $\text{supp}(a_i)$. Hence $\pi(T) \in S_\infty$, so $\pi(T)$ is a multiple of some $\pi(t')$ with $t' \in \mathcal{C}$, i.e. $t' \subseteq T$. By Claim 3.1 t' meets every member of $\mathcal{C}$, so t' is a transversal of $\mathcal{C}$ contained in T ; minimality of T gives $T = t' \in \mathcal{C}$. ■

Step 4: Combinatorial core theorem

Theorem. Let $\mathcal{C}$ be a family of finite subsets of a countable ground set such that (i) (antichain) no member contains another; (ii) (intersecting) any two members intersect; (iii) (self-dual) every minimal transversal of $\mathcal{C}$ is a member of $\mathcal{C}$. Then $\mathcal{C}$ is finite.

Proof. Suppose $\mathcal{C}$ is infinite.

- (a) *Every transversal W of $\mathcal{C}$ contains a minimal transversal.* Consider transversals contained in W ordered by reverse inclusion. If (T_α) is a chain of transversals, then $T := \bigcap_\alpha T_\alpha$ meets every $e \in \mathcal{C}$: indeed $\{e \cap T_\alpha\}$ is a chain (under inclusion) of nonempty subsets of the finite set e ; its distinct members are pairwise comparable subsets of a finite set, so there are finitely many of them, hence the chain has a least member, which is nonempty and equals $e \cap T = \bigcap_\alpha (e \cap T_\alpha)$. So chains have lower bounds; by Zorn's lemma there is a minimal transversal contained in W .

Fix a member $t \in \mathcal{C}$. By (ii) every member meets the finite set t , so

$$\mathcal{C} = \bigsqcup_{\emptyset \neq I \subseteq t} \mathcal{C}_I, \quad \mathcal{C}_I := \{e \in \mathcal{C} : e \cap t = I\},$$

a finite union; hence some $\mathcal{C}_I$ is infinite. Fix such an I .

- (b) *Key observation.* If $g \in \mathcal{C}$ is disjoint from I , then g meets every member of $\mathcal{C}_I$ at a point outside t : $g \cap e \neq \emptyset$ by (ii), and $g \cap e \cap t = (g \cap t) \cap (e \cap t) = (g \cap t) \cap I = \emptyset$ since $g \cap I = \emptyset$.

Fix an enumeration of the countable family $\mathcal{C}$. We inductively build distinct points $z_1, z_2, \dots$ outside t and infinite families $\mathcal{F}_s = \{e \in \mathcal{C}_I : e \supseteq Z_s\}$, $Z_s := \{z_1, \dots, z_s\}$, starting from $\mathcal{F}_0 := \mathcal{C}_I$.

Stage $s + 1$ ($s \geq 0$), with $\mathcal{F}_s$ infinite.

Case A: no member of $\mathcal{C}$ is disjoint from $I \cup Z_s$. Then $I \cup Z_s$ is a transversal; by (a) it contains a minimal transversal T , and $T \in \mathcal{C}$ by (iii). Every $e \in \mathcal{F}_s$ satisfies $e \supseteq I \cup Z_s \supseteq T$, so by (i) $e = T$; hence $\mathcal{F}_s \subseteq \{T\}$, contradicting that $\mathcal{F}_s$ is infinite. Case A is impossible.

Case B: some member is disjoint from $I \cup Z_s$; let g_s be the one of smallest index. Since $g_s \cap I = \emptyset$, observation (b) says g_s meets every $e \in \mathcal{F}_s$ outside t ; as g_s is finite and $\mathcal{F}_s$ infinite, some point $r \in g_s \setminus t$ lies in infinitely many members of $\mathcal{F}_s$. Put $z_{s+1} := r$ (note $r \notin Z_s$ as $g_s \cap Z_s = \emptyset$) and $\mathcal{F}_{s+1} := \{e \in \mathcal{F}_s : z_{s+1} \in e\}$, which is infinite.

Thus the process runs for all $s < \omega$, producing $Z_\omega := \{z_1, z_2, \dots\}$.

- (c) *Every member of $\mathcal{C}$ meets $I \cup Z_\omega$.* Suppose g is disjoint from $I \cup Z_\omega$; then g is disjoint from $I \cup Z_s$ at every stage, hence always eligible. At stage $s + 1$ the least eligible member g_s receives $z_{s+1} \in g_s$, so g_s is ineligible at all later stages; moreover g_{s+1} was eligible at stage $s + 1$ (it is disjoint from $I \cup Z_{s+1} \supseteq I \cup Z_s$), so $\text{index}(g_{s+1}) \geq \text{index}(g_s)$, and $g_{s+1} \neq g_s$ (since $z_{s+1} \in g_s$ but $z_{s+1} \notin g_{s+1}$). Hence $\text{index}(g_s)$ is strictly increasing, therefore unbounded — but $\text{index}(g) \geq \text{index}(g_s)$ for all s , contradiction.

So $I \cup Z_\omega$ is a transversal; by (a) it contains a minimal transversal T , and $T \in \mathcal{C}$ by (iii). Since T is finite and $Z_\omega = \bigcup_s Z_s$, there is a finite s with $T \subseteq I \cup Z_s$. The family $\mathcal{F}_{s+1}$ is infinite; pick $e \in \mathcal{F}_{s+1}$, $e \neq T$. Then $e \supseteq I \cup Z_{s+1} \supseteq I \cup Z_s \supseteq T$, so $e \supsetneq T$, contradicting (i). ■

Conclusion

By Step 3 and Step 4, $\mathcal{C}$ (equivalently $\mathcal{M}$) is finite. By Step 2, there exist positive integers T and L with $a_{n+T} = a_n + L$ for every positive integer n . ■